

Factoring Special Forms in Area and Volume Models

Answer Key

Algebra 1

Quick Answers

1. $(x - 6)(x + 6)$
2. $(x + 5)^2$
3. $(3x - 7)(3x + 7)$
4. $(2x + 3)^2$
5. $2(x - 5)(x + 5)$

6. $x^2 - 49$
 7. $(5x - 4y)(5x + 4y)$
 8. $3(x - 5)^2$
 9. $x(x - 4)(x + 4)$
 10. (a) factored form = $(4x - 3)^2$, —
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What is verified

9 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1

HSA-APR.A.1 – *problem 6*

HSA-SSE.B.3 – *problems 1, 2, 3, 4, 5, 7, 8, 9, 10*

Difficulty 1–4 of 5 across 11 tagged checks.

1. A square sheet of copper has area x^2 square centimetres. A square hole of area 36 square centimetres is punched out of its middle, leaving $x^2 - 36$ square centimetres of metal. Write the remaining area as a product of two binomials.

Solution: Two terms, both perfect squares, separated by a minus sign — this is $a^2 - b^2$ with $a = x$ and $b = 6$, because $6^2 = 36$.

$$x^2 - 36 = x^2 - 6^2 = (x - 6)(x + 6)$$

The two factors are the side lengths, in centimetres, of the rectangle you get by rearranging the leftover metal.

$(x - 6)(x + 6)$

2. A square patio is tiled so that its area is $x^2 + 10x + 25$ square metres. Factor the area completely so that the side length of the patio, in metres, can be read from the answer.

Solution: Three terms. The first and last are perfect squares: $x^2 = (x)^2$ and $25 = 5^2$. Check the middle term against $2ab$: $2 \cdot x \cdot 5 = 10x$, which matches, so the pattern is $a^2 + 2ab + b^2 = (a + b)^2$.

$$x^2 + 10x + 25 = (x + 5)^2$$

The patio is $(x + 5)$ metres on a side.

$$(x + 5)^2$$

3. A square banner measures $3x$ centimetres on each side. A square logo measuring 7 centimetres on each side is cut out of it, leaving $9x^2 - 49$ square centimetres of fabric. Write the remaining area as a product of two binomials.

Solution: Write each term as a square first: $9x^2 = (3x)^2$ and $49 = 7^2$. Now the difference-of-squares pattern applies with $a = 3x$, $b = 7$.

$$9x^2 - 49 = (3x)^2 - 7^2 = (3x - 7)(3x + 7)$$

$$(3x - 7)(3x + 7)$$

4. The floor of a square storage shed has area $4x^2 + 12x + 9$ square feet. Factor the area completely so that the side length of the shed, in feet, can be read from the answer.

Solution: $4x^2 = (2x)^2$ and $9 = 3^2$, so try $a = 2x$, $b = 3$. The middle-term test decides it: $2ab = 2(2x)(3) = 12x$, which is exactly the middle term.

$$4x^2 + 12x + 9 = (2x + 3)^2$$

The shed floor is $(2x + 3)$ feet on a side.

$$(2x + 3)^2$$

5. A rectangular deck has area $2x^2 - 50$ square feet. Factor the expression completely. Take out the common factor first.

Solution: Both terms share a factor of 2. Pull it out before looking for a pattern — otherwise neither term is a perfect square and the difference-of-squares pattern is hidden.

$$\begin{aligned} 2x^2 - 50 &= 2(x^2 - 25) \\ &= 2(x - 5)(x + 5) \end{aligned}$$

$$2(x - 5)(x + 5)$$

6. A rectangular window pane measures $(x + 7)$ inches along one edge and $(x - 7)$ inches along the other. Write its area, in square inches, in standard form with no parentheses.

Solution: Area is length times width. Multiply the binomials; the two middle products cancel, which is exactly why the answer is a difference of squares.

$$\begin{aligned} (x + 7)(x - 7) &= x^2 - 7x + 7x - 49 \\ &= x^2 - 49 \end{aligned}$$

$$x^2 - 49$$

7. Two square vegetable plots lie side by side. The larger plot has side $5x$ metres and the smaller plot has side $4y$ metres, so the difference of their areas is $25x^2 - 16y^2$ square metres. Write that difference as a product of two binomials.

Solution: Each term is a perfect square in its own variable: $25x^2 = (5x)^2$ and $16y^2 = (4y)^2$. The pattern does not care that two different letters appear.

$$25x^2 - 16y^2 = (5x)^2 - (4y)^2 = (5x - 4y)(5x + 4y)$$

$$(5x - 4y)(5x + 4y)$$

8. A square rug and its carpet border together cover $3x^2 - 30x + 75$ square feet. Factor the expression completely: take out the greatest common factor first, then look at what is left.

Solution: The greatest common factor of $3x^2$, $30x$ and 75 is 3 . After it comes out, the leftover trinomial is a perfect square: $x^2 = (x)^2$, $25 = 5^2$, and $2 \cdot x \cdot 5 = 10x$ matches the middle term.

$$\begin{aligned} 3x^2 - 30x + 75 &= 3(x^2 - 10x + 25) \\ &= 3(x - 5)^2 \end{aligned}$$

$$\boxed{3(x - 5)^2}$$

9. The volume of a shipping crate, in cubic inches, is modelled by $x^3 - 16x$, where x is a length in inches. Factor the expression completely so that the three dimensions of the crate can be read from the answer.

Solution: The common factor is x . Take it out first; what remains is a difference of squares, so the factoring is not finished until that is broken up too.

$$\begin{aligned} x^3 - 16x &= x(x^2 - 16) \\ &= x(x - 4)(x + 4) \end{aligned}$$

The three dimensions, in inches, are x , $(x - 4)$ and $(x + 4)$.

$$\boxed{x(x - 4)(x + 4)}$$

10. A square window pane has area $16x^2 - 24x + 9$ square inches. (a) Factor the area completely. (b) Explain how the numbers 16, 24 and 9 tell you this expression is a perfect-square trinomial rather than a difference of squares.

Solution (a): $16x^2 = (4x)^2$ and $9 = 3^2$, so the candidate pattern is $a^2 - 2ab + b^2$ with $a = 4x$, $b = 3$. Test the middle term: $2ab = 2(4x)(3) = 24x$, and the sign is negative, so the factored form uses a minus.

$$\boxed{(4x - 3)^2}$$

(b) — instructor-judged. A full answer must make three points: the expression has *three* terms, while a difference of squares has only two; the first and last terms are the perfect squares of $4x$ and 3 ; and the middle term $-24x$ is twice the product of $4x$ and 3 , which is the signature of $a^2 - 2ab + b^2$. An answer that stops at “both ends are squares” has not checked the middle term and should not get full credit — $16x^2 - 20x + 9$ has square ends and does not factor this way.